

Enhancing XR Accessibility through Anthropometrically Diverse Biomechanical Simulation

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ABSTRACT

Biomechanical user simulations can complement efforts to enhance XR accessibility by enabling rapid evaluation of alternative design choices across a wider range of user body types and capabilities. However, it is unclear to what extent current biomechanical simulation policies generalize to different body shapes. To address this gap, we investigate how biomechanical models for mid-air selection perform under anticipated diversity in user anthropometry. Using a musculoskeletal simulator with reinforcement learning (RL), we analyzed how variations in arm length influence model performance. The results reveal that existing RL policies are applicable only to the narrow range of anthropometric conditions for which they were trained, with significant performance deterioration outside this range. Our study highlights the importance of developing flexible and general biomechanical models capable of representing anthropometrically diverse users.

Index Terms: XR, Accessibility, Biomechanical Simulation, Reinforcement Learning.

1 INTRODUCTION

As XR technologies become more widely adopted, it will be increasingly important to ensure accessibility for users with different physiological characteristics [2]. User simulation can potentially complement other more human-centered methods to evaluate and enhance the accessibility of XR. However, many traditional approaches to user modeling, such as Fitts' law [4], rely on simplified predictive formulas which abstract from the complexity of human body movement. This means that they are fundamentally limited in ensuring physiological diversity. As more people of all sizes and abilities become involved with XR, there is a need for more advanced methods.

Due to advances in both hardware and software, biomechanical user simulations have become a promising approach to overcome these difficulties [9]. Instead of relying on simplified predictive formula or human-subject experiments, simulation-based methods integrate detailed dynamic models of the user's body to create accurate user representations [10, 1, 5, 7]. Simulations can then be run to explore the consequences of various design choices and to predict performance bottlenecks.

A common strategy for building these simulators involves utilizing RL to control a biomechanical model, either using simplified torque actuators [3] or virtual muscles [8]. RL has shown promise in approximating human-like decision-making and movement patterns [1, 5, 7], learning to complete tasks in realistic virtual environments.

Historically, these biomechanical models have been trained on a single reference model – most current upper extremity models inherit the body physiology of the original Holzbaur model [6], where bone lengths were set to match a “50th percentile male (170 cm

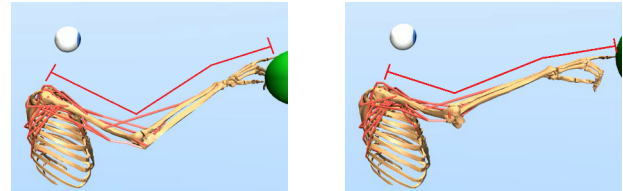


Figure 1: **Left:** Simulated pointing movement with default arm length (77.5cm). **Right:** Adjusting the arm length (here: 69.4cm) considerably reduces simulation quality, often leading to failure.

tall)”. There is, however, potential to represent a more diverse population and deliver high quality predictions across a wide range of users by adjusting physiological parameters, such as arm length. These models could then be used to personalize interfaces that would be too time-consuming or ethically challenging to systematically evaluate directly with human participants. In this work, we focus exclusively on the influence of arm length given its relevance to mid-air interaction in XR, but other relevant physiological parameters such as muscle strength and movement constraints could be similarly explored. Given the focus on arm length, this study also evaluates the accuracy of the Meta Movement SDK in estimating user arm lengths, exploring its potential as a method for directly capturing user anthropometric data.

2 METHODOLOGY

The objective of this research is to evaluate the current state-of-the-art in biomechanical simulators by assessing user diversity in terms of varying arm length. By doing so, this study aims to uncover limitations in current simulators and highlight opportunities for improvement.

Focusing on hand-based mid-air interaction, we consider a range of arm lengths to represent anthropometrically diverse users and examine how these variations impact the performance. Using anthropometric data from NASA-STD-3001 [11], we identified the 5th and 95th percentile arm lengths. For our biomechanical simulator, we use the “User-in-the-Box” model from Ikkala et al. [7], which integrates a musculoskeletal model with visual and proprioceptive perception and RL to generate task-specific muscle control policies. This simulator is recognized for its ability to develop muscle-actuated control policies. The default arm length within the model is 77.5 cm (see Figure 1).

To systematically explore how arm length influences performance, we sampled 20 additional arm lengths between the 5th and 95th percentile, in addition to the default setting. For each arm length, we simulated a pointing task where the simulated user attempts to move the virtual fingertip onto a specified target efficiently and accurately. We used the same reward function proposed by Ikkala et al. [7], consisting of a distance penalty, neural effort costs, and a one-time bonus for successfully hitting a target.

To explore how anthropometrically diverse user simulations could be leveraged within a potential XR application scenario, we also tested the ability of the Meta Movement SDK to reliably estimate user arm lengths when a user assumes 10 different body poses.

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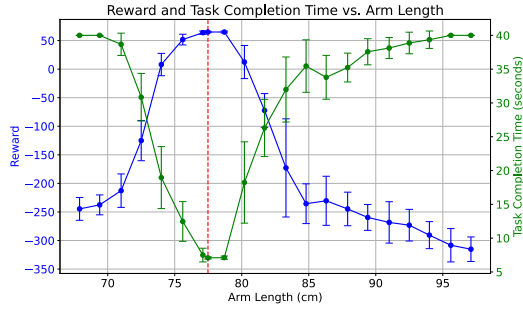


Figure 2: Reward and completion time with varying arm lengths.

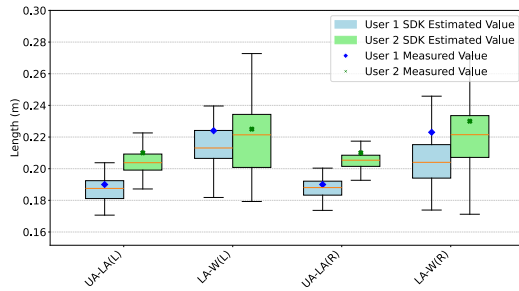


Figure 3: Meta Movement SDK arm length estimation.

3 RESULTS

3.1 Effect of Varying Arm Length

Our simulations revealed a clear relationship between arm length and the quality of simulated performance in the pointing task. Using the default arm length from the User-in-the-Box model as a midpoint (77.5 cm), we evaluated two key metrics: reward and task completion time.

Figure 2 plots the reward and task completion time at various arm lengths. The red dotted vertical line shows the default user arm length. Figure 2 highlights that the reward peaks around 77–79 cm, aligning closely with the default arm length. Deviations beyond this range result in significant drops in reward. Similarly, the RL policy achieves the fastest completion times with minimal variation within the 77–79 cm interval. Conversely, both shorter and longer arm lengths lead to substantially increased completion time. Note that the simulated completion time is limited to a maximum of 40 seconds. Pilot studies with a range of different arm lengths showed that all considered target positions can be reached in principle, suggesting that the observed decline in performance stems from limitations in the model rather than an inherent user inability to complete the task.

In summary, our analysis identifies an optimal arm length range of 77–79 cm. Deviating more than 2 cm from this default setting results in performance deterioration, indicating the model's inability to generalize to variation in anthropometric parameters.

3.2 Quality of Arm Length Estimates

The box plot in Figure 3 compares arm segment lengths obtained from the Meta Movement SDK for two users. The relevant arm segments are the upper arm to lower arm (UA-LA) and the lower arm to wrist (LA-W) for both right (R) and left (L) arms. In most cases, the dots fall within the inter-quartile ranges, demonstrating that the Movement SDK provides consistent and reasonably accurate estimates for arm lengths. Additionally, the plot illustrates that

measurements for the same segments on the right and left sides are similar, suggesting the ability to capture symmetry in human anatomy.

4 CONCLUSION

This study highlights the current shortcomings of state-of-the-art biomechanical models for the task of effectively simulating performance across a broader range of user anthropometry. Analysis of the effect of varying arm length revealed a fairly narrow optimal range for task efficiency and stability. In particular, deviating more than 2 cm from the default arm length led to an observable reduction in performance. This poor generalizability of the simulator outside of the trained anthropometric conditions highlights its deficiencies for real-world user-specific applications. Nevertheless, our analysis demonstrated the reliability of the Movement SDK in capturing realistic and symmetrical anthropometric data. This suggests there are opportunities for optimizing user-specific simulations in XR environments if the generalizability of these biomechanical models can be improved. Addressing this limitation is essential to increase the accuracy of simulation-based predictions, and unlock the potential of this complementary method for enhancing XR accessibility.

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